

Dr. Babasaheb Ambedkar Technological University, Lonere

Final Year Electrical

Summer Examination 2019

Subject: High Voltage Engineering (EE803)

Semester: VIII

Date: 20/05/2019

Duration: 3hour

Marks: 70

Note: Figure on right indicate max marks. Solve sequentially. All questions are compulsory.

- Q 1. Attempt the following (12M)**
- a) Derive the Townsends current growth equation. (6)
 - b) Explain various factors influencing breakdown in gases (6)
- OR**
- c) What is time lag? Why it occurs? What are the different factors which affects on it? (6)
- Q.2. Attempt the following (12M)**
- a) What are the different methods and means for purification of liquid dielectrics (6)
 - b) Explain breakdown due to treeing and tracking in solid insulating materials under electrical stress (6)
- OR**
- c) What is meant by composite dielectrics? Explain its properties (6)
- Q.3. Attempt the following (12M)**
- a) What are the mechanisms by which lightning strokes are developed and induce overvoltage on overhead power line? (6)
 - b) What is meant by insulation coordination? How are the protective devices chosen for optimal insulation level in power system? (6)
- Q.4. Attempt the following (12M)**
- a) Explain with diagram different types of rectifier circuits used for producing High DC voltages (6)
 - b) Explain different schemes for cascade connection of transformer for producing very high ac voltages <https://www.batuonline.com> (6)
- OR**
- c) Describe the circuit arrangement for producing impulse current waveform in laboratories (6)
- Q.5. Attempt the following (12M)**
- a) What are the different methods of dc voltage measurement? Explain high ohmic series resistance with micro ammeter (6)
 - b) Explain resistance potential divider for dc voltage measurement (6)

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- Q.6. Attempt the following** (10M)
- a) Explain the terms
- i. Withstand voltage
 - ii. 50% flashover voltage (5)
 - iii. 100% flashover voltage
 - iv. Creepage distance
 - v. Wet and dry power frequency test
- b) What are the different power frequency tests done on insulators? Mention the procedure for testing (5)

*******All the Very Best*******

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